



The Effect of Peer Academic Support on the Motivation of Students of the Uika Bogor Digital Business Study Program

Febrina Rahmadiani Putri, Miftahul Jannah, Pinkan Prilia Mahmudah, Salma Difa Aristawidya, Asti Marlina *

Digital Business Study Program, Faculty of Economics and Business, Universitas Ibn Khaldun Bogor
Jl. Sholeh Iskandar No.Km.02, RT.01/RW.010, Kedungbadak, Kec. Tanah Sereal, Kota Bogor, Jawa Barat
16162, Indonesia

E-mail: asti@uika-bogor.ac.id

Article Information	Abstrak
History of the article: Accepted: March 2024 Corrected: April 2024 Publish: June 2024 Keywords: Peer Academic Support, Achievement Motivation, Students	This study aims to analyze the influence of peer academic support peers to the motivation to excel students of the Digital Business Study Program Ibn Khaldun University Bogor. Peer academic support is seen as an external factor that plays a role in encouraging students to achieve optimal academic achievement. This study uses quantitative approach with survey methods. Data is collected through questionnaire distributed to 74 respondents of the Business Study Program students Digital. The data analysis technique used is Partial Least Squares–Structural Equation Modeling (PLS-SEM) with the help of software SmartPLS. The analysis was carried out through the evaluation stage of the measurement model (outer model), evaluation of structural models (inner models), and hypothesis testing for assess the significance of the relationship between variables. The results of the study show that peer academic support has a positive and significant effect on students' motivation to excel. These findings indicate that the higher the academic support that students receive from their peers, the higher the motivation of students to excel. This research is expected to make a theoretical and practical contribution in development of a supportive academic environment to increase motivation student achievements.

Introduction

Education has a strategic role in shaping quality and competitive human resources. Through higher education, students are not only equipped with academic knowledge, but also critical thinking skills, professional attitudes, and social skills needed to face the dynamics of an increasingly competitive world of work. Universities are required to produce graduates who are adaptive to technological developments and digitalization, so that the learning process is no longer oriented to knowledge transfer alone, but encourages students to become active, independent, and responsible learners for their learning process (Wicaksono et al., 2025).

In line with the student-centered learning paradigm, students are positioned as the main actors in the learning process through active involvement in discussions, problem-solving, and collaborative work. This approach allows for a more meaningful construction of knowledge and supports the development of cognitive, affective, and social aspects in a balanced manner (Fani et al., 2024). Thus, a student's academic success is determined not only by intellectual ability, but also by the psychological and social factors that accompany the learning process.

One of the psychological factors that plays an important role in the academic success of students is the motivation to excel. Achievement motivation encourages individuals to set academic goals, strive to achieve the best results, and endure various learning challenges. McClelland defines achievement motivation as an individual's drive to achieve a certain standard of excellence through maximum effort (Ilham et al., 2022). Students with high achievement motivation tend to show diligence, responsibility, and consistency in completing academic tasks. However, in practice, not all students have the same level of motivation to excel, so some students experience a decrease in enthusiasm for learning, lack of confidence, or difficulty in facing academic demands.

The difference in the level of motivation for achievement is influenced by various factors, both from within the student and from the surrounding environment. Internal factors include learning interests,

academic goals, and self-efficacy, while external factors include the learning environment and social support. In the context of higher education, the academic social environment plays an important role because the learning process takes place in dynamic social interaction between students. Positive social interaction can create a conducive, supportive, and motivating learning environment for students to be actively involved in academic activities (Wicaksono et al., 2025).

Relationships with peers are an important part of a student's academic social environment. Peers are often the main source of sharing lecture information, learning experiences, and emotional support during their studies. The similarity of academic experience and emotional closeness make peer interaction have a significant influence on students' attitudes, learning behaviors, and motivation. In this context, the support provided between students is known as peer academic support.

Peer academic support is a form of social support that focuses on learning activities and academic achievement, provided by peers at the same academic level. This support includes academic and psychosocial assistance that helps students face the demands of lectures more effectively (Rejeki et al., 2023). Forms of peer academic support can be in the form of group discussions, learning together, sharing learning resources, providing academic feedback, and emotional support when facing learning difficulties.

Based on the study of social support in an academic context, peer academic support is classified into several main forms, namely instrumental support, informational support, emotional support, and assessment support (Rejeki et al., 2023). Instrumental support is related to direct assistance in academic activities, informational support includes the delivery of information relevant to lectures, emotional support in the form of empathy and attention, while assessment support is in the form of feedback that helps students understand and evaluate their academic abilities. The four forms of support are interrelated in forming a positive learning experience.

A number of previous studies have shown that peer support has an effect on students' learning motivation and academic adjustment. A supportive social environment helps students feel more comfortable, confident, and able to manage academic pressure better (Shanti & Setiadi, 2021). This support has also been proven to have a positive and significant relationship with student achievement motivation, where the existence of supportive peers encourages students to continue to strive to achieve optimal academic results (Ilham et al., 2022). In addition, peer social support contributes to increasing students' academic resilience, so that students are better able to survive, not give up easily, and find solutions adaptively when facing various difficulties in the learning process (Dewi et al., 2025).

The role of peer academic support is becoming increasingly relevant in the context of digital learning. Universities today require students to learn independently and collaboratively through discussions, group work, and technology-based project completion. The Digital Business Study Program, which emphasizes the integration of technology, creativity, and collaboration, presents complex academic demands and has the potential to cause pressure if not balanced with a supportive learning environment. In these conditions, academic support from peers is an important source of motivation and emotional stability for students.

Although various studies have examined the relationship between social support and student learning motivation, most studies are still general and have not specifically highlighted peer academic support. In addition, previous research subjects generally came from other study programs, so studies that specifically analyzed the influence of peer academic support on the motivation to excel students of the Digital Business Study Program were still relatively limited. In fact, the characteristics of learning in the field of Digital Business have their own uniqueness that is different from other fields of study.

Based on this description, this study is focused on analyzing the influence of peer academic support on the motivation of students to excel in the Digital Business Study Program. In particular, this study aims to examine whether peer academic support has an effect on students' motivation to excel, the extent to which this support contributes to increasing achievement motivation, and identifies the aspects of peer academic support that have the most dominant influence on the achievement motivation of students of the Digital Business Study Program. The results of the research are expected to make a theoretical contribution to the development of learning motivation studies as well as practical implications for collaborative and supportive learning management in higher education.

Research Methodology

This study uses a quantitative approach with a survey method to analyze the influence of peer academic support on the motivation of students to excel in the Digital Business Study Program, Ibn Khaldun University (UIKA) Bogor. The quantitative approach was chosen because this study aims to empirically test the relationship between variables through numerical measurement and statistical analysis. The research population includes all students of the UIKA Bogor Digital Business Study Program which totals 280 students from various generations. The sample was determined using the probability sampling technique with

the simple random sampling method, so that each member of the population has the same opportunity to become a respondent. The number of samples was determined using the Slovin formula with a margin of error of 10%, so that 74 respondents were obtained used in the main study.

Data collection was carried out using a closed questionnaire compiled based on indicators of research variables and measured using a five-level Likert scale, ranging from 1 (strongly disagree) to 5 (strongly agree). Before the questionnaire was used in the main study, an initial test (pilot test) was carried out on 30 respondents to ensure the quality of the research instruments. This stage aims to test the validity and reliability of the indicators used in measuring the construct of peer academic support and student achievement motivation.

The convergent validity test was carried out by evaluating the outer loading value of each indicator. According to Hair et al. (2014), the indicator is declared to meet the convergent validity if it has an outer loading value of ≥ 0.70 .

Table 1 : Results of Convergent Validity Test (Outer Loading) – Initial Test of 30 Samples

	Academic Support	Motivation to Excel.
X1	0.834	
X2	0.859	
X3	0.848	
X4	0.882	
X5	0.918	
X6	0.894	
X7	0.884	
X8	0.910	
Y1		0.814
Y2		0.787
Y3		0.834

Based on Table 1, all indicators in both variables have an outer loading value above 0.70. This shows that each indicator is able to represent the construct it measures well, so that all indicators are declared to be valid in a convergent manner and suitable for use at the next stage of research.

Furthermore, the reliability and validity tests of the construct were carried out by evaluating the values of Cronbach's Alpha, Composite Reliability, and Average Variance Extracted (AVE). Hair et al. (2014) stated that the construct is declared reliable if the value of Cronbach's Alpha and Composite Reliability is greater than 0.70, and meets the validity of the construct if the AVE value is greater than 0.50.

Table 2 : Results of Construct Reliability and Validity Test – Initial Test 30 Samples

	Cronbach's Alpha	Composite reliability (rho_a)	Composite reliability (rho_c)	Average variance extracted (AVE)
Academic Support	0.958	0.961	0.965	0.773
Motivation Achievement	0.935	0.939	0.947	0.690

The results in Table 2 show that both constructs have excellent Cronbach's Alpha and Composite Reliability values, as well as AVE values that have exceeded the required minimum limits. Thus, it can be concluded that all constructs in this study are reliable and valid, so that the research instrument is declared suitable for use in the main study.

The variables used in this study consist of independent variables and dependent variables. The independent variable is the academic support of peers, while the dependent variable is the motivation of students to achieve. The operational definition of the variable is compiled based on indicators relevant to the theoretical foundation and summarized in Table 3.

Table 3 : Research Variables and Operational Definition

	Variabel		Indicator
	Item Code	Statement	
Peer Academic Support (X)	Instrumental support	X1, X6	Direct assistance in understanding the material and tasks
	Support Informational	X2, X3	Providing information related to materials, assignments, and academics
	Emotional support	X4, X7	Encouragement and enthusiasm in facing lectures
	Assessment support	X5, X8	Input and evaluation of academic abilities
Motivation to Excel (Y)	Achievement orientation	Y1, Y2	Targets and desires to achieve achievements
	Perseverance in learning	Y3, Y6	Consistency Facing academic difficulties
	Challenges and efforts	Y4, Y5	Efforts to improve academic ability
	Performance satisfaction	Y7, Y8	Top satisfaction Academic Achievement

Data analysis in the main study was carried out using the Partial Least Squares– Structural Equation Modeling (PLS-SEM) method with the help of SmartPLS software. This method is used because it is able to analyze the relationship between latent variables simultaneously and is suitable for research with a relatively limited sample size. The analysis stages include the evaluation of the measurement model (outer model) to ensure the validity and reliability of the construct, as well as the evaluation of the structural model (inner model) to assess the causal relationship between variables and test the research hypothesis based on the resulting significance (Hair et al., 2014).

Results

This study involved 74 student respondents of the Digital Business Study Program, Ibn Khaldun University Bogor. Respondent characteristics are presented to provide an overview of the profile of the research sample.

Table 4: Characteristics of Respondents by Gender

Gender	Quantity	Percentage%
Male	35	43,3%
Women	39	52,7%
Total	74	100%

Based on Table 4, the respondents were dominated by female students, which shows that the participation of female students in this study is higher than that of male students.

Table 5: Characteristics of Respondents by Force

Force	Quantity	Percentage%
2022	7	9,5%
2023	41	55,4%
2024	23	31,1%
2025	3	4%
Total	74	100%

Furthermore, the characteristics of respondents by class (Table 5) show that respondents come from various active generations, with the dominance of 41 students in the class of 2023 (55.4%). The class of 2024 students totaled 23 people (31.1%), followed by the class of 2022 as many as 7 people (9.5%), and the class of 2025 as many as 3 people (4%).

The evaluation of the measurement model was carried out to assess the quality of the indicators in reflecting the latent constructs being studied. The first test is a convergent validity test which is carried out by looking at the outer loading value of each indicator against its construct. The convergent validity test aims to assess the extent to which the indicator is able to reflect the measured construct.

Table 6 : Results of Convergent Validity Test (Outer Loading)

	Academic Support	Motivation to Excel.
X1	0.842	
X2	0.842	
X3	0.870	
X4	0.903	
X5	0.916	
X6	0.911	
X7	0.872	
X8	0.916	
Y1		0.853
Y2		0.847
Y3		0.875
Y4		0.873
Y5		0.839
Y6		0.903
Y7		0.901
Y8		0.870

All indicators in the variables of peer academic support and achievement motivation have an outer loading value above the minimum limit of 0.70. This shows that each indicator is able to explain latent constructs strongly and consistently. Thus, all indicators are declared to be convergently valid and suitable for use in follow-up testing.

Furthermore, the reliability and validity of the construct are tested to ensure internal consistency and the ability of the construct to explain the variables.

Table 7 : Results of Construct Reliability and Validity Test

	Cronbach's Alpha	Composite reliability (rho_a)	Composite reliability (rho_c)	Average variance extracted (AVE)
Academic Support	0.960	0.961	0.966	0.782
Motivation to Achieve	0.954	0.956	0.962	0.758

Based on the results of the reliability and validity tests of the construct (Table 7), the Cronbach's Alpha and Composite Reliability values for both variables were above the threshold of 0.70, while the Average value The Variance Extracted (AVE) of each variable exceeds 0.50. These results show that the constructs of peer academic support and achievement motivation have high reliability and excellent construct validity.

The validity of the discriminator is tested to ensure that each construct has a clear conceptual difference from each other.

Table 8 : HTMT Test Results

Variabel	Academic Support	Motivation to Excel.
Academic Support	-	-
Motivation to Excel.	0.831	-

Based on the results of the Heterotrait–Monotrait Ratio (HTMT) test in Table 8, the HTMT value between academic support and achievement motivation construct is 0.831, which is below the maximum limit of 0.90. This shows that both constructs have a good degree of discrimination.

Table 9: Fornell-Larcker Test Results

Variabel	Academic Support	Motivation to Excel.
Academic Support	0.884	-
Motivation to Excel.	0.801	0.870

In addition, the results of the Fornell–Larcker test (Table 9) show that the root value of AVE in each construct is greater than the correlation between constructs. These results confirm that the validity of the discriminant has been met, so that each construct can be empirically distinguished.

The next model evaluation is the evaluation of the structural model. This evaluation was conducted to test the relationship between latent constructs in the research model. The results of the path coefficient analysis (Table 10) show that peer academic support has a positive and significant effect on students' motivation to achieve, with a path coefficient value of 0.801, a T-statistics value of 13.870, and a p-value of 0.000.

Table 10 : Path Coefficient Results

	Original sample (O)	Sample mean (M)	Standard deviation (STDEV)	T statistics (O/STDEV)	P values
Support Academic → Motivation Achievement	0.801	0.801	0.058	13.870	0.000

These results show that an increase in academic support from peers is significantly followed by an increase in student achievement motivation. Thus, the research hypothesis that states the influence of peer academic support on students' motivation for achievement is accepted.

The strength of the structural model is also analyzed through the R-square value. Based on Table 11, the R- square value for the achievement motivation variable is 0.642.

Table 11 : R-Square Value

Variable endogenous	R-Square
Motivation to Excel.	0.642

This value shows that peer academic support is able to explain 64.2% of the variation in student achievement motivation, while the remaining 35.8% is influenced by other factors outside the research model. These values indicate that the model has a strong clarity in explaining endogenous variables.

In addition, effect size testing through the f-square value was carried out to determine the magnitude of the influence of exogenous variables on endogenous variables. The results of the f-square test in Table 12 show a value of 1.793 for the relationship of peer academic support to achievement motivation.

Table 12 : F-Square Value

Hubungan	F-Square
Academic Support → Motivation for Achievement	1.793

This value far exceeds the limit of 0.35, which shows that peer academic support has a large effect size on students' motivation to excel.

An analysis of the outer loading of bootstrap results was carried out to identify the most dominant indicators in reflecting each construct. Based on Table 13, all indicators have p-values < 0.05, which indicates that each indicator is statistically significant.

Table 13 : Outer Loading Bootstrap Results (summary of key indicators)

Indicator	Or	M	T-Statistics	P-Values
X5	0.916	0.913	37.869	0.000
X8	0.916	0.913	31.811	0.000
Y6	0.903	0.900	26.808	0.000
Y7	0.901	0.896	28.135	0.000

In the peer academic support variable, the X5 indicators (peer provide input or suggestions that help improve academic ability) and X8 (peer support in achieving academic goals) have the highest outer loading and sample mean values. This shows that academic evaluation-based support and encouragement for goal achievement are the most powerful aspects in shaping students' perceptions of peer academic support.

In the achievement motivation variable, indicators Y6 (perseverance in completing tasks despite facing obstacles) and Y7 (satisfaction when achieving higher achievements) showed the highest outer loading and sample mean values. These findings indicate that students' motivation for achievement is most strongly reflected in resilience in facing academic challenges and a sense of satisfaction with achievements. Thus, the motivation for student achievement is not only reflected in the existence of academic targets, but also from persistence and internal satisfaction for the success achieved.

Discussion of Research Results

The results of the study show that peer academic support plays a significant role in increasing the motivation to excel students of the Digital Business Study Program. Evaluative, informational, and emotional support has proven to be an important factor in encouraging students to keep trying, set academic goals, and survive the pressure of lectures.

The indicator with the greatest contribution comes from the assessment support aspect which shows that students are very helpful when they get feedback, suggestions, and reinforcement from their peers. These findings confirm that a collaborative academic environment not only serves as a means of information exchange, but also as a source of motivation and psychological reinforcement.

In the achievement motivation variable, the most dominant indicator was related to perseverance and achievement satisfaction, which reflected that highly motivated students were less likely to give up easily and had an internal drive to achieve better results. These findings are in line with McClelland's (Ilham et al., 2022) view that achievement motivation is characterized by a strong desire to achieve standards of excellence.

Overall, the results of this study reinforce previous findings that peer academic support is a strategic external factor in shaping student achievement motivation, especially in the context of collaboration-based learning such as in the Digital Business Study Program.

Conclusion

This study aims to analyze the influence of peer academic support on the motivation of students of the Digital Business Study Program. Based on the results of analysis and discussion, this study shows that peer academic support has a positive and significant effect on student achievement motivation. The support provided through academic interaction between students plays a role in encouraging the setting of learning targets, increasing perseverance in completing assignments, and helping students survive various academic challenges. These results show a compatibility between the research objectives formulated in the introduction and the empirical evidence obtained.

In addition, the evaluation of the measurement model and structural model showed that the constructs and indicators used met the criteria of validity and reliability, so that they were able to adequately represent the concepts of peer academic support and achievement motivation. Substantively, this study emphasizes the importance of a supportive academic social environment in supporting student achievement motivation, especially in the context of collaboration-based learning such as in the Digital Business Study Program. The practical implications of this study lead to the need to strengthen a collaborative and supportive learning culture between students as part of a strategy to improve the quality of learning. Further research is recommended to examine other relevant variables, expand the context and characteristics of respondents, and use a more diverse methodological approach to enrich understanding of the factors that influence students' motivation to achieve.

References

- Fakultas, D. I., & Komunikasi, I. (2025). HUBUNGAN EFIKASI DIRI DAN DUKUNGAN SOSIAL TEMAN SEBAYA DENGAN RESILIENSI AKADEMIK PADA MAHASISWA. 9(1), 47–54.
- Fani, A., Nur, M., Nurdin, H., & Djalal, N. M. (2024). Hubungan Antara Dukungan Sosial Teman Sebaya dan Motivasi Berprestasi pada Mahasiswa Berorganisasi. 3(4).
- Ilham, M., Wibawanti, I., & Sovitriana, R. (2019). Dukungan Sosial Teman Sebaya Dengan Motivasi Berprestasi Pada Mahasiswa Di Fakultas Psikologi Universitas Persada Indonesia Y . A . I. 2(2), 43–50.
- Rejeki, Y. F., Rahayu, S., Sebaya, T., & Belajar, M. (2023). Dukungan sosial teman sebaya dengan motivasi belajar pada mahasiswa. XVII, 102–110.
- Shanti, T. I., & Setiadi, B. (2021). Friends ' Support , Motivation to Learn , Emotional Adjustment , and Academic Performance among Indonesian First-Year Students. 48, 67–82. <https://doi.org/10.22146/jpsi.34060>
- Wicaksono, T. N., Utami, N. I., & Belajar, M. (2025). Dukungan sosial teman sebaya dengan motivasi belajar pada mahasiswa. 5(4), 4191–4196.